During the negative half cycle of the supply, diodes D2 and D4 conduct in series while D1 and D3 are reverse-biased. The current flows through the load as shown in Figure 4.14.

The bridge rectifier is also called the full wave rectifier as it produces an output pulse for each half-cycle of the input sine wave. The input and output waveforms are shown in figure 4.15.

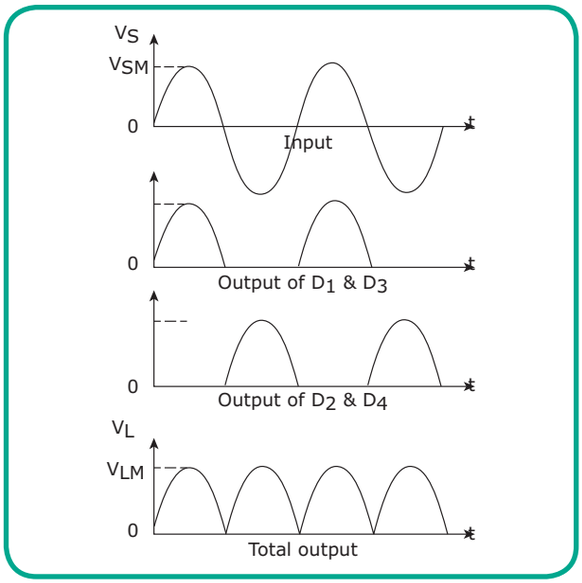


Figure 4.15 Input and output waveforms of bridge rectifier

$$Output\ of\ the\ full\ wave\ rectifier = \frac{2E_{max}}{\pi}$$

$$\therefore\ The\ Average\ DC\ Voltage = 0.637\,E_{max}$$

$$Ripple\ Factors = 2 \times f$$

$$Efficiency = 81.2\%$$

4.7 FILTER CIRCUITS

To remove the AC components or filter-out them in a rectifier circuit, a filter circuit is used. A filter circuit is a device to remove the AC components of the rectifier output (pulsating DC – it is also called ripples), but allows the DC components to reach the load.

A filter circuit consists of passive elements, i.e., inductors, capacitors, resistors and their combinations. Some of the important filters are given below.

1. Inductor or Choke filter
2. Capacitor filter
3. RC filter
4. Inductor-Capacitor filter (LC filter)
5. π or pie filter or CLC filter

4.7.1 Inductors Filter or Choke Filter

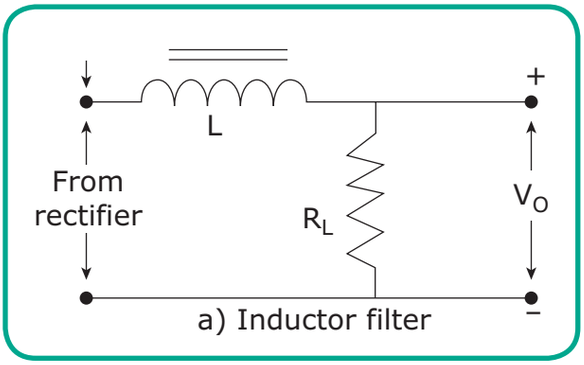


Figure 4.16 Inductor filter circuit

Figure 4.16 shows the inductor filter circuit. This type of filter consists of an inductor L inserted between the rectifier and the load resistance R_L. The rectifier output contains AC components as well as DC components. When the output passes through the inductor it offers a high resistance to the AC components and no resistance to the DC components. Therefore, an AC component of the rectifier is blocked and only DC component reaches at the load.

4.7.2 Capacitor Filter

Figure 4.17 shows the capacitor filter circuit. In this filter, an electrolytic capacitor is used as a reservoir capacitor, because it acts as a temporary storage for the power supply output current. During the rise of voltage cycle, it gets charged and this charge is supply to the load during the fall in the voltage cycle.

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